Spring 2026: ME759 Final Project Proposal

Quick intro:

- Name: Yupeng Zhang
- Email: zhang2599@wisc.edu
- Home department: Math
- Status: Undergraduate student/MS Student/PhD Student. PhD
- Name of your teammate (if applicable):
- Link to your repo for the Final Project: https://github.com/yzhang-math/Fused_Attn_Kernel.git
    - NOTES:
        - This may or may not be your ME759 repo (it's not the same if you work on a team).
        - Make the repo private for now. It'll be made public on the day you submit your project

Project Title: Fused Attention Kernel
Link to git repo for project: https://github.com/yzhang-math/Fused_Attn_Kernel.git

Problem statement: I will implement a CUDA-based "Fused Attention" kernel. The goal is to compute the Attention mechanism (Query, Key, Value matrix multiplications and the Softmax operation) entirely within the GPU's Shared Memory (SRAM) without writing intermediate $N \times N$ attention scores to Global Memory. To maximize compute throughput, the kernel will utilize mixed-precision math (FP16 inputs/outputs with FP32 accumulation) via hardware Tensor Cores.

Motivation/Rationale: To understand how we speed up transformers.

Explain how you contemplate going about it: I will use pure CUDA and the nvcuda::wmma API. I will avoid high-level libraries like cuBLAS for the core implementation to ensure I learn the low-level mechanics. My approach is phased:

- Write a CPU C++ reference of Attention and the "Online Softmax" algorithm for mathematical verification.
- Write a naive, unfused CUDA implementation as a performance baseline.
- Develop the highly optimized Fused Attention kernel utilizing Shared Memory tiling and Tensor Cores.
- Profile the kernels using NVIDIA Nsight Compute.

ME759 aspects the proposed work draws on:

- GPU architecture and memory hierarchy (Global vs. Shared Memory).
- Thread block and warp-level synchronization.
- Shared memory optimization (resolving bank conflicts via data padding).
- Tensor Core programming using the Warp Matrix Multiply Accumulate (wmma) API.
- Performance profiling and Roofline Analysis.

Deliverables:

- C++/CUDA source code containing the CPU reference, the naive baseline kernel, and the optimized Tensor Core Fused Kernel.
- A final technical report detailing the mathematical approach to Online Softmax, the shared memory padding strategies used, and the performance scaling results.

How you will demonstrate what you accomplished: I will demonstrate success by presenting a comparative performance analysis using data extracted via Nsight Compute. I will show a Roofline Model chart comparing my fused kernel against the naive baseline across different architectures (Ada Lovelace, Ampere, Hopper). The defining metric of success will be a quantifiable reduction in Global Memory read/write traffic (GB/s) and an increase in computational throughput (TFLOPS) achieved by keeping data in SRAM and utilizing Tensor Cores.

Other remarks: none